AWS Networking, Compute and Storage

AWS foundations, volume 2 · Version 2026.1

> Sample foundation document written for testing a domain-expert RAG. AWS services, limits and model availability change often; confirm details in the official AWS documentation.

1. VPC fundamentals

A Virtual Private Cloud (VPC) is an isolated network in one region, spanning its Availability Zones. Subnets live in a single Availability Zone. A **public subnet** has a route to an internet gateway; a **private subnet** does not, and reaches the internet (if at all) through a NAT gateway in a public subnet. Place load balancers in public subnets and application servers and databases in private subnets across at least two Availability Zones.

2. Security groups and network ACLs

	Security group	Network ACL
Applies to	Elastic network interfaces (instances, load balancers, endpoints)	Whole subnets
State	Stateful: return traffic is allowed automatically	Stateless: return traffic needs its own rule, including ephemeral ports
Rules	Allow rules only	Allow and deny rules
Evaluation	All rules evaluated together	Rules evaluated in number order; first match wins
Typical use	Primary, fine-grained control; reference other security groups by ID	Coarse subnet-level blocks, such as denying a known bad range

3. Private connectivity

VPC endpoints let resources reach AWS services without going over the internet.

Gateway endpoints exist for S3 and DynamoDB and are added to route tables at no hourly charge. **Interface endpoints** (AWS PrivateLink) place elastic network interfaces in your subnets for many other services and for services offered by other accounts. Endpoint policies can restrict which buckets or actions are reachable. For hybrid connectivity use Site-to-Site VPN or AWS Direct Connect; for many VPCs use AWS Transit Gateway.

4. Compute choices

Service	Best for	Notes
Amazon EC2	Full control, GPUs, long-running services	Choose instance families by workload: general, compute, memory, accelerated
AWS Lambda	Event-driven functions and APIs	Maximum run time 15 minutes; memory configurable from 128 MB to 10,240 MB; ephemeral storage up to 10 GB
Amazon ECS / AWS Fargate	Containers without managing a cluster (Fargate) or with EC2 capacity	Simpler than Kubernetes for most teams
Amazon EKS	Managed Kubernetes	When you need the Kubernetes ecosystem

4.1 EC2 purchasing options

Option	When to use
On-Demand	Unpredictable or short workloads
Savings Plans / Reserved Instances	Steady usage committed for 1 or 3 years, at a large discount
Spot Instances	Fault-tolerant, interruptible work such as batch and CI; up to about 90% cheaper; AWS gives a two-minute interruption notice
Dedicated Hosts	Licensing or compliance needs for physical isolation

5. Load balancing and scaling

The Application Load Balancer works at layer 7 (HTTP/HTTPS) with path and host routing, and integrates with AWS WAF. The Network Load Balancer works at layer 4

(TCP/UDP/TLS) for very high throughput and static IPs. The Gateway Load Balancer inserts third-party appliances such as firewalls. EC2 Auto Scaling groups replace unhealthy instances and scale on metrics such as CPU or request count per target.

6. Amazon S3

S3 is designed for 99.999999999% (eleven nines) durability. New objects are encrypted at rest by default with SSE-S3; use SSE-KMS when you need control over who can decrypt and an audit trail of key use.

Storage class	Use
S3 Standard	Frequently accessed data
S3 Intelligent-Tiering	Unknown or changing access patterns; moves objects between tiers automatically
S3 Standard-IA / One Zone-IA	Infrequent access; One Zone-IA keeps data in a single Availability Zone
S3 Glacier Instant Retrieval	Archive data that still needs millisecond access
S3 Glacier Flexible Retrieval / Deep Archive	Long-term archives retrieved in minutes to hours

6.1 Securing a bucket with customer data

1. Keep S3 Block Public Access on at account and bucket level.
2. Disable ACLs (Object Ownership: bucket owner enforced) and use bucket policies.
3. Require TLS with an `aws:SecureTransport` deny statement.
4. Encrypt with SSE-KMS using a customer managed key, and restrict the key policy.
5. Enable versioning, and Object Lock where records must not be altered.
6. Restrict access to specific roles and, where possible, to your VPC endpoint.
7. Turn on CloudTrail data events or server access logging, and Amazon Macie to find sensitive data.
8. Use lifecycle rules to delete data at the end of its retention period.

7. Databases

Service	Type	Notes

Amazon RDS	Managed relational (PostgreSQL, MySQL, others)	Multi-AZ deployments for automatic failover; read replicas for scale
Amazon Aurora	Cloud-native MySQL/ PostgreSQL compatible	Storage replicated across three Availability Zones; Aurora Serverless for variable load
Amazon DynamoDB	Key-value and document NoSQL	Single-digit millisecond latency; on-demand or provisioned capacity; global tables
Amazon ElastiCache	In-memory cache (Redis-compatible, Memcached)	Session stores and hot data
Amazon OpenSearch Service	Search and analytics, including vector search	Serverless option available